International Journal of Recent Advances in Multidisciplinary Topics
Volume X, Issue X, Month XXXX
https://www.ijramt.com | ISSN (Online): 2582-7839

Enhanced Natural Language Interaction Platform with Offensive Content Filtering

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***Abstract*: Real-time messaging platforms impose strict inference latency constraints that large transformer models routinely exceed, whereas regular expression-based filters cannot detect context-dependent toxicity such as indirect threats and implicit insults. This study introduces a tiered inference architecture that resolves this constraint without ensemble overhead. Explicit profanity is intercepted by a client-side, word-boundary-aware regex filter operating under 1 ms, offloading trivial cases before server involvement. Contextual toxicity detection was delegated to a server-side fine-tuned DistilBERT classifier trained on the MetaHate dataset (1,101,157 samples) under class-imbalanced conditions (3.72:1 ratio) using weighted cross-entropy loss. At a calibrated decision threshold of θ = 0.625, the classifier yields 89.94% accuracy, 77.37% toxic-class F1, and a ROC-AUC of 95.06%, with a PR-AUC of 85.63%, a more discriminative metric under class imbalance. End-to-end message latency measures 80–140 ms under concurrent load, within the 250 ms threshold established for synchronous chats. When benchmarked against seven baseline architectures spanning classical ML, RNN, and gradient boosting variants under identical experimental conditions, DistilBERT yielded the strongest accuracy-latency ratio, outperforming the next best validated architecture (BiLSTM) by 3.15 percentage points in accuracy and 5.40 points in toxic-class F1. The architecture was validated using a deployed messaging prototype.**

***Keywords:* Content Moderation, DistilBERT, Natural Language Processing, Real-Time Systems, Tiered Inference, Toxic Content Detection**

1. INTRODUCTION

The widespread use of online communication platforms, such as instant messaging apps and multiplayer games, has increased the frequency of cyberbullying and harassment. These malicious acts harm personal safety and mental health. Recent World Health Organization studies have highlighted that 15% of adolescents have experienced cyberbullying. Existing moderation systems on popular social platforms use a reactive approach. They act after the psychological impact on the recipient has already occurred. Platforms such as Twitter/X and Reddit rely on post-hoc reviews or community reports. Although this approach works for specific cases, it is rendered useless in instant messaging applications.

Some proactive measures can easily be bypassed by intentional lexical errors such as character substitution, intentional spacing, or homoglyph encoding using visually identical Unicode characters from alternate scripts. These methods operate within the latency constraints of real-time messaging but often fail to detect context-dependent toxicity, including insults and sarcasm.

Machine learning and deep learning classifiers offer stronger detection capabilities but introduce different restrictions. Machine learning classifiers are fast in inference but lack contextual understanding. Deep learning models, except transformer architectures, are not accurate enough for this task. In addition, the inference time of transformer models exceeds the latency requirements of 100–250 ms, including processing and network latency.

To address this gap, this study uses a different approach. Primarily, we filter the obvious and explicit profanity using a list of predetermined profane words, and if any such word is present in the text, we block the message on the user side of the application. If it passes through the primary filter, the message goes through the secondary filter, which then classifies the text contextually. Hence, profanity is blocked along with the contextually toxic messages.

2. LITERATURE SURVEY

A. Evolution of Content Moderation Approaches

In the early stages of content moderation, such systems could only identify exact words in the text using regex-based string matching. This is a simple and quick approach that can be performed on the client side, but it requires being explicit about the words, making the algorithm easy to bypass using space and character substitution. When the text contains words like 'Scunthorpe' (a town in England), the algorithm can give false positives because the substring matches the target phrase. These structural limitations motivated a transition to statistical classification methods.

B. Classical Machine Learning Methods

Subsequent approaches applied statistical classifiers such as logistic regression, support vector machines (SVM), and Naive Bayes to sentence-level toxicity classification. These models represent text as bag-of-words or TF-IDF feature vectors, discarding word order entirely. This representation precludes sensitivity to semantic context: the sentences "you are dead,

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your mother saw what you did" (non-toxic) and an identical surface form used as a direct threat carry the same feature vector. Similarly, historically contextualized terms and words with distinct meanings across time periods or communities produce systematic false positives when evaluated purely on lexical co-occurrence [2, 3]. While classical models offer high throughput (exceeding 50,000 samples/s for SVM and logistic regression), their inability to encode sequential or contextual structures limits toxic-class F1 to the low-to-mid 60s on benchmark datasets, as confirmed by our evaluation in Section 4.

C. Deep Learning Architectures

Limitations of classical ML led researchers toward deep learning solutions. ML algorithms treat sentences as bag-of-words and ignore word order, but in language, sequence matters, 'tiger hunts man' and 'man hunts tiger' are not the same. Recurrent Neural Networks (RNNs) attempted to solve this sequence problem by maintaining a hidden state across the sentence [16], allowing the model to understand sentiment shifts.

However, RNNs have limited memory and introduce vanishing/exploding gradients, making them notoriously difficult to train. To address this, researchers introduced Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) [14, 16]. These architectures use internal gates to store context, capturing dependencies in paragraphs as well as sentence-level context. Bidirectional LSTMs (BiLSTM) process text in both forward and backward directions [7, 16], allowing the network to understand a word's meaning in past and future contexts [15]. While BiLSTMs consistently outperformed simple RNNs, LSTMs, and GRUs in toxicity detection benchmarks [16], they remain computationally heavy and their sequential nature renders them too slow for high-volume, real-time applications [14].

D. Transformer-Based Approaches

A paradigm shift occurred with the introduction of the Transformer architecture. Unlike RNNs, which read words sequentially, Transformers use a "self-attention" mechanism to process entire sequences simultaneously [1, 4]. This helps Bidirectional Encoder Representations from Transformers (BERT) understand the relationships between words by considering every word in relation to every other word [4]. Pre-training on massive datasets, followed by fine-tuning on toxicity data, has made BERT a strong contender for text classification, sentiment detection, and more [5, 17, 18].

BERT and its variants consistently achieve high accuracy on hate speech and toxic comment datasets [1, 4], excelling at detecting subtle toxicity, indirect insults, threats, and sarcasm. These architectures support transfer learning, enabling fine-tuning for racial bias mitigation [17] and detection of offensive content on social media [5]. Multilingual BERT (mBERT) extends these capabilities to cross-language detection, which is critical for global platforms [1, 7, 15]. Although BERT's high accuracy and context awareness incur a larger memory footprint and inference time [13], making deployment difficult in latency-sensitive environments, researchers developed DistilBERT through knowledge distillation. DistilBERT retains 97% of BERT's performance with only 60% of its parameters while running 60% faster.

E. Real-Time Chat Systems

Applications such as YouTube, Telegram, and WhatsApp operate under strict latency constraints [9, 10]. Even a one-second delay can disrupt the flow of conversation [11, 12]. Although large language models offer superior reasoning capabilities, they are too slow and expensive for synchronous chat [8]. Scalability and cost are bottlenecks, processing over 100,000 messages per second is a real-world requirement; therefore, most real-time systems still use hash-based filtering or simple classifiers to maintain throughput.

F. Research Gap and Proposed Solution

A clear divide exists in the current literature. Academic research often prioritizes model accuracy over inference speed and memory footprint, frequently proposing large transformer ensembles that require heavy GPU resources [1]. Production environments prioritize low latency, cost-efficiency, scalability, and memory, thus relying on simple regex filters or keyword blocking. There is a scarcity of models that combine strong performance with practical efficiency. Real-time chat systems demand 100–250 ms response times, which pure BERT models exceed. This research fills this gap by proposing a hybrid approach combining traditional word filtering with a fine-tuned DistilBERT model for contextual analysis.

3. PROPOSED SYSTEM

The proposed system operates in two stages. The first stage runs on the client side: it checks each outgoing message against a curated dictionary of prohibited terms using word-boundary-aware regular expression matching along with case normalization. The second stage is a fine-tuned DistilBERT model (66.5 million parameters) used as the primary classifier. DistilBERT classifies messages in 30 ms on the P100 GPU and can detect indirect insults, implicit threats, and sarcasm that lexical filters miss.

Combined with a network round-trip latency of 40–100 ms for 50 concurrent users and 10 ms of internal server overhead, the total end-to-end latency ranges from 80–140 ms, within the sub-250 ms limit required for synchronous chat [11, 12]. Tail latency under higher concurrency loads was not measured, which is noted as a limitation in Section 6.

A. Dataset and Exploratory Analysis

For this research, the source of data for training was MetaHate by IRLab-UDC on huggingface.co [20]. The dataset is in TSV (tab-separated values) format, containing 1,226,202 social media posts (1,101,157 after cleaning) with fields Text and Label (toxic/non-toxic), and serves as a benchmark for real-world content moderation due to its diverse linguistic styles.

The dataset exhibits moderate class imbalance, with the non-toxic class comprising 78.81% of the data (867,869 samples),

causing a class imbalance ratio of 3.72:1. Exploratory analysis shows toxic messages are shorter (mean length: 31 words) than non-toxic messages (mean length: 49 words). A stratified train-validation split resulted in 935,983 training samples (737,784 non-toxic, 198,199 toxic) and 165,174 validation samples (130,181 non-toxic, 34,993 toxic). Text is tokenized using DistilBERT's WordPiece tokenizer (vocabulary: 30,522) with [CLS] and [SEP] tokens, truncated or padded to 128 tokens (covering more than 99% of data samples without truncation). Attention masks are used to distinguish between text and padding. To address class imbalance, balanced class weights [0.634, 2.360] are computed using scikit-learn, penalizing toxic misclassification 3.72 times more heavily than non-toxic records.

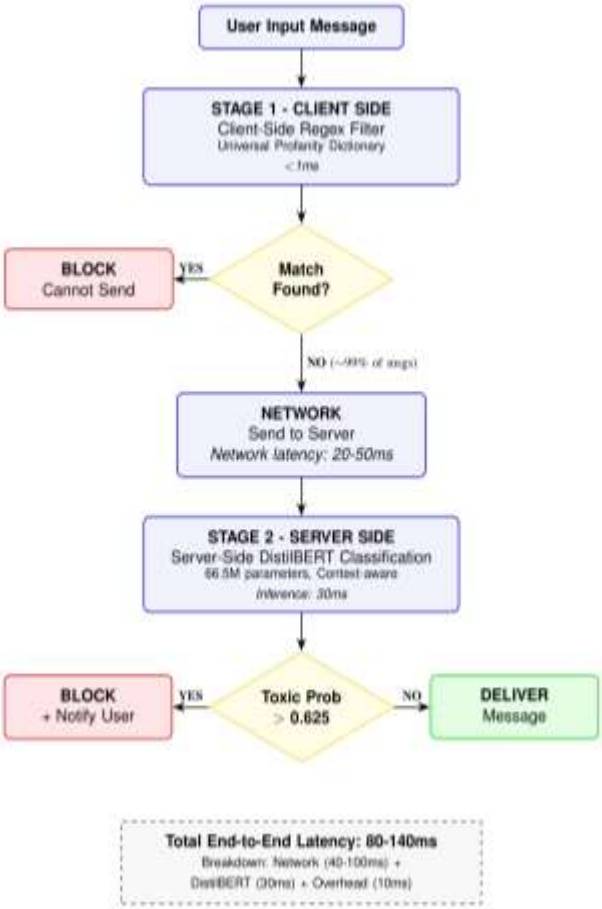

Fig. 1. Proposed two-stage content filtering pipeline with client-side regex and server-side DistilBERT classifier.

B. Model Architecture

Our approach utilized DistilBERT-base-uncased, a 6-layer transformer encoder with 66,362,880 parameters, 768-dimensional hidden states, and 12 attention heads, achieving 60% faster inference than the BERT model while retaining 97% of its accuracy. Multi-head self-attention processes all tokens simultaneously, enabling the model to understand sarcasm (e.g., 'I hope you have the day you deserve') and implicit threats.

The classification layer was replaced with a custom head (197,378 parameters) comprising: Dropout (p = 0.3), dense layer (768 to 256 dimensions), GELU activation, Dropout (p = 0.27), and a final dense layer (256 to 2 classes). The output bias was initialized with class priors to improve detection of minority classes. Total model size is 66,560,258 parameters (253.9 MB FP32). Model outputs binary logits are converted using the softmax activation function.

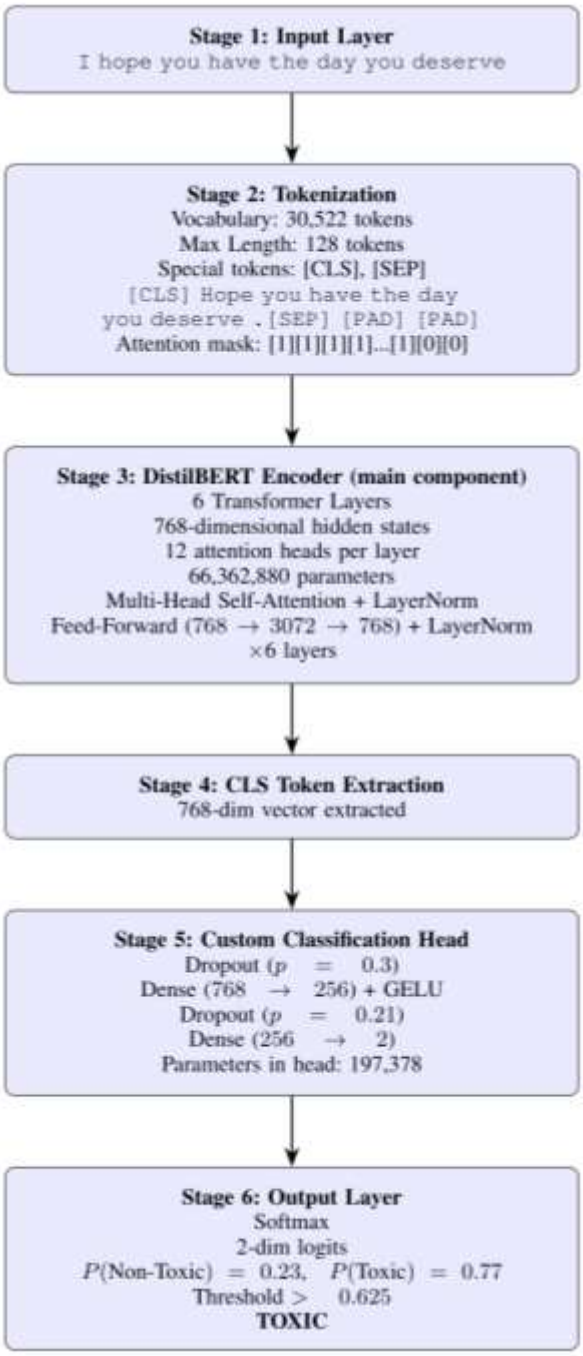

Fig. 2. DistilBERT model architecture showing tokenization, transformer layers, and classification head.

C. Training Configuration

The optimizer of choice was AdamW (learning rate = 3×10^{-5}, weight decay = 0.01, eps = 1×10^{-8}). The loss function was Weighted Cross-Entropy Loss with class weights [0.634, 2.360]. A batch size of 64 with gradient accumulation over 2 steps (effective batch size = 128) was used to fully utilize the P100-PCIE GPU. The learning rate follows a linear warmup over 2,193 steps (6% of 36,560 total) followed by linear decay. Dropout regularization was applied with probability p = 0.3. Gradient clipping was set to max norm = 1.0. Training time was 467.0 minutes for 5 epochs using automatic mixed precision (AMP). The best checkpoint (Epoch 4) achieved a weighted F1 score of 89.61% on validation.

Post-training, threshold values from 0.25 to 0.75 were tested; a threshold of 0.625 gave the maximum weighted F1 of 90.11%, achieving a toxic F1 of 77.37%, balancing precision at 73.89% and recall at 81.19%.

D. Experimental Setup

The experimental setup integrates the following tools: PyTorch 2.9.0 as the core deep learning framework, supported by the Transformers 5.2.0 library for loading and fine-tuning DistilBERT; NumPy and Pandas for data manipulation; FastAPI for serving the DistilBERT model via a REST API; and Kaggle as the development platform providing GPU resources. A fixed seed value of 42 was set across all experiments to ensure reproducibility.

4. RESULTS AND DISCUSSION

A. Evaluation Metrics

The DistilBERT model was evaluated on 165,174 validation samples using accuracy, weighted F1, binary F1 (toxic class), precision, recall, ROC-AUC, and PR-AUC. At the optimal threshold ($\theta = 0.625$), the confusion matrix yielded: True Negatives: 120,134; False Positives: 10,047; False Negatives: 6,580; True Positives: 28,413.

The model achieved 89.94% accuracy and a weighted F1 score of 90.11%. For the toxic class, the binary F1 score was 77.37%, balancing precision (73.89%) and recall (81.19%), indicating strong performance despite class imbalance. The ROC-AUC of 95.06% confirms strong class separability. The PR-AUC of 85.63% provides a more informative measure under the 3.72:1 class imbalance. The complete metrics are presented in Table 2.

B. Comparison with Baseline Models

Table 1 compares DistilBERT against seven baseline models trained on the MetaHate dataset. Among all evaluated models, DistilBERT achieves the highest accuracy (89.94%), toxic F1 (77.37%), ROC-AUC (95.06%), and PR-AUC (85.63%). It outperforms the next best validated architecture, BiLSTM, by 3.15 percentage points in accuracy and 5.40 points in toxic-class F1.

Classical machine learning models including SVM, Logistic Regression, XGBoost, and Naive Bayes offer massive throughput (213,000–10,372,000 samples/s), but perform poorly at classifying messages. XGBoost achieves only 56% toxic F1 and misses over half of toxic messages (43% recall). This occurs because these models treat text as a bag of words, ignoring word order and context.

Table 2
DistilBERT Performance Metrics

Metric	Value
Accuracy	89.94%
Weighted F1	90.11%
Toxic F1 (Binary)	77.37%
Toxic Precision	73.89%
Toxic Recall	81.19%
Non-Toxic Precision	94.81%
Non-Toxic Recall	92.29%
Non-Toxic F1	93.53%
ROC-AUC	95.06%
PR-AUC	85.63%
Training Time	467.0 min
Inference (batch)	86 samples/s
Inference (single)	11.6 ms/message

C. Error Analysis

To understand the limitations of DistilBERT, we analyzed false positives (n = 10,047) and false negatives (n = 6,580) from its predictions. For false positives, two primary patterns emerged: (1) sarcasm and irony, messages like "Oh great, another brilliant idea" (toxic probability 0.762) trigger false alarms due to negative words while missing the ironic intent; (2) quoted content, Reports of abuse sometimes cause misclassifications because the model flags the quoted toxic language without recognizing the reporting context.

False negatives consist of three types: (1) indirect threats that are only threatening in specific contexts (e.g., "People like you don't last here more than a week"); (2) short text or emojis where context is absent, which may include insulting nicknames or coded terms; (3) toxicity hidden in hyperlinks or

mentions. The false positive rate was relatively high at 7.72%, which may adversely affect speakers of African American Vernacular English, members of communities who reclaim specific terminology, and non-native speakers of English.

Table 2
Comparison with Baseline Models

Model	Architecture	Accuracy	F1-Toxic	Tox Recall	Train Time	Inference (samples/s)
DistilBERT	Transformer	89.94%	77.37%	81.19%	467.0 min	86
BiLSTM	RNN	86.79%	71.97%	80.02%	142.9 min	6,511
GRU	RNN	85.36%	71.19%	85.40%	198 min	20,920
XGBoost	Boosting	85.34%	56.43%	44.80%	178.7 s	415,075
Naive Bayes	Classical	82.97%	55.85%	50.83%	0.3 s	5,292,141
Logistic Reg.	Classical	81.84%	65.47%	81.26%	12.1 s	9,119,673
LightGBM	Boosting	81.76%	64.07%	76.74%	230.8 s	213,534
Linear SVM	Classical	81.63%	65.24%	81.40%	65.8 s	10,372,257

D. Production Deployment Considerations

The decision threshold $\theta = 0.625$ was selected through calibration over the range [0.25, 0.75], maximizing the F1 score on the validation set. This requires higher model confidence to flag text as toxic, reducing false positives at the cost of a modest reduction in toxic recall. This is appropriate for synchronous chats, where incorrectly blocked messages carry an immediate cost to the user experience. For deployment, users can report falsely flagged messages (false positives) and missed toxic messages (false negatives) for human review.

5. CONCLUSION

This paper addresses the latency-accuracy trade-off in real-time content moderation with two stages: a client-side regex filter handling explicit profanity, and a server-side fine-tuned DistilBERT classifier handling context-dependent toxicity. The system achieves 89.94% accuracy on MetaHate, 77.37% toxic F1, and end-to-end latency of 80–140 ms fast enough for real-time chat. These results indicate that production-grade content moderation does not require large-ensemble configurations or billion-parameter models. A lightweight design keeps costs low and still captures most toxicity. The system is not without limitations; sarcasm and indirect threats remain challenging failure modes. Nevertheless, the proposed pipeline meets the latency requirements while outperforming all classical and RNN-based approaches evaluated in this study.

6. FUTURE WORK

Several directions emerge from the limitations identified in this study. Multilingual support is a necessary extension, as the current model processes only English, limiting its applicability on global platforms. Sarcasm detection requires improvement in the direction of analyzing the context of the entire conversation rather than a single message. We will also explore model quantization and compression techniques to make the model deployable without dedicated GPU infrastructure. Finally, we wish to explore multimodal content moderation, including images, audio, and video.

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